A Topological Approach to the Simple Descriptions of Convex Hulls of Sets Defined by Three Quadrics

MEGA 2024

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Problem Description

Fix symmetric matrices $Q_1, Q_2, Q_3 \in \mathbb{R}^{(n+1)\times(n+1)}$ and consider

$$S = \left\{ x \in \mathbb{R}^n \;\middle|\; \begin{bmatrix} x^\top & 1 \end{bmatrix} Q_i \begin{bmatrix} x^\top & 1 \end{bmatrix}^\top \leq 0,\ i = 1, 2, 3 \right\}.$$

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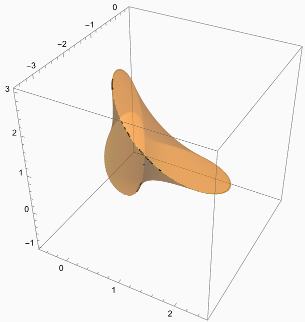
Running Example:[1]

$$Q_1 = \begin{bmatrix} 25 & 0 & -32 & 0 \\ 0 & 25 & 0 & 24 \\ -32 & 0 & 6 & 0 \\ 0 & 24 & 0 & 6 \end{bmatrix},\ Q_2 = \begin{bmatrix} 0 & 0 & 12 & 0 \\ 0 & 0 & 0 & 16 \\ 12 & 0 & 4 & 0 \\ 0 & 16 & 0 & 4 \end{bmatrix}, \text{ and}$$

$$Q_3 = \begin{bmatrix} 0 & 0 & 0 & -60 \\ 0 & 0 & 10 & 0 \\ 0 & 10 & 0 & 0 \\ -60 & 0 & 0 & 0 \end{bmatrix}$$



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Problem Description

We consider *aggregations* $\lambda \in \mathbb{R}^3_+$ and associated supersets of S:

$$S \subseteq S_\lambda := \left\{ x \in \mathbb{R}^n \;\middle|\; \begin{bmatrix} x^\top & 1 \end{bmatrix}^\top \left(\sum_{i=1}^{3} \lambda_i Q_i \right) \begin{bmatrix} x^\top & 1 \end{bmatrix}^\top \leq 0 \right\}$$

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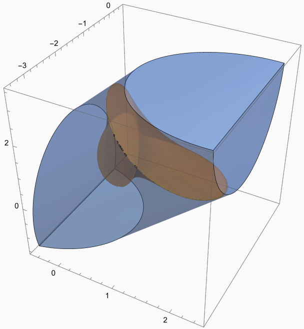

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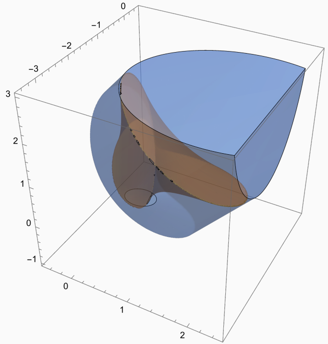

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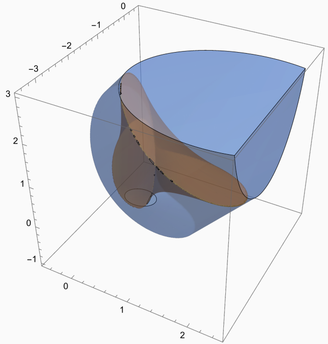

- $Q_{\lambda^{(1)}} = (1 - 0.57)Q_1 + 0.57Q_2$
- $Q_{\lambda^{(2)}} = (1 - 0.34)Q_1 + 0.34Q_3$
- λ is a *good aggregation* if Q_λ has ≤ 1 negative eigenvalue and $\operatorname{conv}(S) \subseteq S_\lambda$.

Problem Description

Question: When is there $\Lambda_1 \subseteq \mathbb{R}^3_+$ with $\operatorname{conv}(S) = \bigcap_{\lambda \in \Lambda_1} S_\lambda$?

[2] *Dey, Santanu S., Muñoz, Gonzalo, and Serrano, Felipe. "On obtaining the convex hull of quadratic inequalities via aggregations". SIAM J. Optim. 2022.*

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Previous Work:[2,3] Sufficient condition: Given $H \subseteq \mathbb{R}^{n+1}$ obtained as homogenization of a hyperplane valid on S, there exists $\lambda \in \mathbb{R}^3_+$ such that $Q_\lambda|_H \succ 0$.

This is implied if there exists $\mu \in \mathbb{R}^3$ with $Q_\mu \succ 0$. In this case, we say *PDLC* holds.

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Our Approach: Study the *spectral curve*

$$\mathcal{V}(g(\lambda)) = \mathcal{V}\left(\det\left(\sum_{i=1}^{3} \lambda_i Q_i\right)\right) \subseteq \mathbb{RP}^2$$

using Agrachev-Lerario spectral sequence[4]

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Main Theorem

Theorem (Convex Hull From Aggregations)

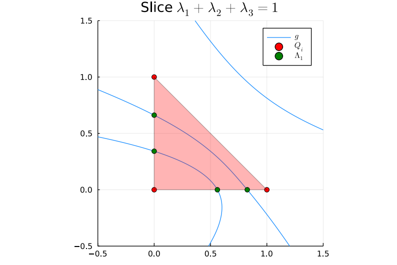
Assume that $\operatorname{int}(S) \neq \emptyset$*, and that* $\{x \in \mathbb{R}^{n+1} \mid x^\top Q_i x \leq 0,\ i \in [3],\ x_{n+1} = 0\} = \{0\}$*. Suppose that* $\mathcal{V}_{\mathbb{R}}(f_1^h, f_2^h, f_3^h)$ *is empty and the spectral curve is smooth and nonempty.*

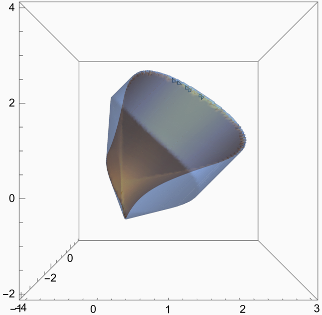
1. *If* $n \geq 3$*, then the spectral curve* g *is hyperbolic. Let* $\mathcal{P} \subset \mathbb{R}^3$ *be the hyperbolicity cone of* g *consisting of either positive semidefinite matrices* Q_λ*, or matrices* Q_λ *with exactly two negative eigenvalues. If no non-zero aggregation lies in* $\mathcal{P}$ *then there are finitely many good aggregations* $\lambda^{(1)}, \lambda^{(2)}, \ldots, \lambda^{(k)}$ *such that* $\operatorname{conv}(S) = \bigcap_{i=1}^k S_{\lambda^{(i)}}$*.*
2. *If* $n = 2$ *and* g *is hyperbolic, then (1) still applies. If* g *is not hyperbolic, then there is a (possibly infinite) subset* $\Lambda_1 \subseteq \mathbb{R}^n_+$ *of good aggregations such that* $\operatorname{conv}(S) = \bigcap_{\lambda \in \Lambda_1} S_\lambda$*.*

Nonexample

In the running example, $(1,0,0)$ is in the hyperbolicity cone of g and

$$\operatorname{conv}(S) \subsetneq \bigcap_{\lambda} S_\lambda$$

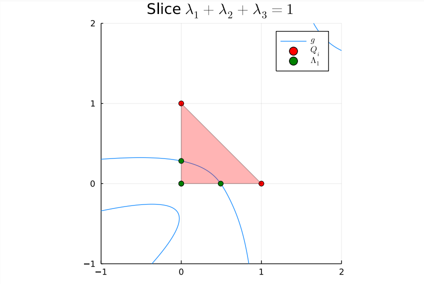


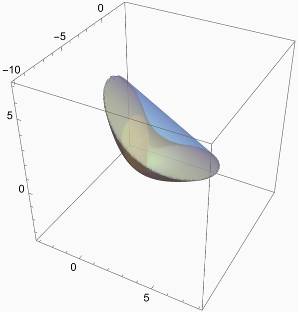


Example

Modify running example so that Q_1 is not contained in $\mathcal{P}$:

$$\tilde{Q}_1 = (0.3Q_1 + 0.4Q_2 + 0.3Q_3)\,, \quad \tilde{Q}_2 = Q_2, \quad \tilde{Q}_3 = Q_3$$





Agrachev-Lerario Spectral Sequence

Let

- $K \subseteq \mathbb{R}^3$ a polyhedral cone, $f_i^h(x) = x^\top Q_i x$ the quadratic form associated to Q_i, and

$$X = \left\{ [x] \in \mathbb{RP}^n \;\middle|\; \begin{bmatrix} f_1^h([x]) & f_2^h([x]) & f_3^h([x]) \end{bmatrix}^\top \in K \right\}$$

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- K° polar dual of K and

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Then, there exists a spectral sequence (E_r, d_r) converging to $H_{n-*}(X)$ such that d_2 is computable and

$$E_2^{ij} \cong \begin{cases} H^{i-1}(\Omega^{j+1}), & i \geq 2,\ \Omega^{j+1} \neq \emptyset \\ H^0(\Omega^{j+1})/\mathbb{Z}_2, & i = 1,\ \Omega^{j+1} \neq \emptyset \\ \mathbb{Z}_2, & i = 0,\ \Omega^{j+1} = \emptyset \\ 0, & \text{otherwise} \end{cases}. \tag{1}$$

Empty Varieties

Theorem (Hyperbolicity of the Spectral Curve)

Let f_1^h, f_2^h, f_3^h be the quadratic forms $f_i^h(x) = x^\top Q_i x$. Suppose that the spectral curve $\mathcal{V}_{\mathbb{R}}(g)$ is smooth and $n \neq 2$. Then, the real projective variety $\mathcal{V}_{\mathbb{R}}(f_1^h, f_2^h, f_3^h) \subseteq \mathbb{RP}^n$ is empty if and only if one of the following holds:

1. *The polynomial g is hyperbolic, and there is $\mu \in \mathbb{R}^3$ such that Q_μ has n positive eigenvalues*
2. *$n = 3$ and the spectral curve $\mathcal{V}_{\mathbb{R}}(g)$ is empty.*

When $n = 2$, the spectral curve is not smooth if the variety $\mathcal{V}_{\mathbb{R}}(f_1^h, f_2^h, f_3^h) \subseteq \mathbb{RP}^n$ is nonempty.

Remark

Previously known: $n = 3$ case[5], "only if" statement[6] for $n \geq 4$.

[5] *Plaumann, Daniel, Sturmfels, Bernd, and Vinzant, Cynthia. "Quartic curves and their bitangents". J. Symbolic Comput. 2011.*

[6] *Agrachëv, A. A. "Homology of the intersections of real quadrics". Dokl. Akad. Nauk SSSR. 1988.*

Hyperbolic Polynomials

Definition: A homogeneous polynomial $h \in \mathbb{R}[x, y, z]$ is *hyperbolic* with respect to $e \in \mathbb{R}^3$ if $h(te + a) \in \mathbb{R}[t]$ is real rooted for any $a \in \mathbb{R}^3$.

- If h is smooth, $\mathcal{V}_{\mathbb{R}}(h)$ has maximally nested ovals
- The connected component of e in $\mathbb{R}^3 \setminus \widehat{\mathcal{V}_{\mathbb{R}}(h)}$ is a convex cone

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Example:[7] The determinant from the running example:

$$g(x, y, z) = \det \begin{bmatrix} 25x & 0 & -32x + 12y & -60z \\ 0 & 25x & 10z & 24x + 16y \\ -32x + 12y & 10z & 6x + 4y & 0 \\ -60z & 24x + 16y & 0 & 6x + 4y \end{bmatrix}$$

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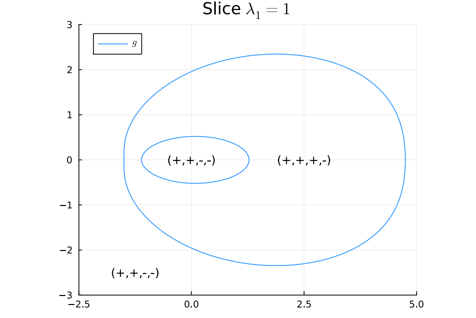
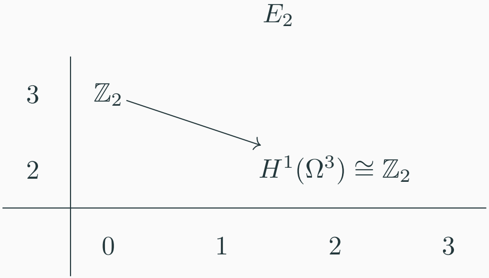
Example

If $K = \{0\} \subseteq \mathbb{R}^3$, then X is a projective variety. The spectral curve and E_2 page for the running example are shown below:





Key Point

Proposition

If $\mathcal{V}_{\mathbb{R}}(f_1^h, f_2^h, f_3^h) = \emptyset$, *then a system of inequalities*

$$\begin{bmatrix} x^\top & 1 \end{bmatrix} \left(\sum_{i=1}^{3} \lambda_i^{(j)} Q_i \right) \begin{bmatrix} x^\top & 1 \end{bmatrix}^\top \leq 0 \text{ for } j = 1, 2, \ldots, m$$

has no solution if and only if the set Ω^n *of combinations* Q_μ *for* $\mu \in \operatorname{cone}(\lambda^{(1)}, \lambda^{(2)}, \ldots, \lambda^{(m)})$ *with at least* n *positive eigenvalues either*

- *contains a positive definite matrix*
- *has* $\dim_{\mathbb{Z}_2} H^1(\Omega^n) = 1$.

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Takeaway

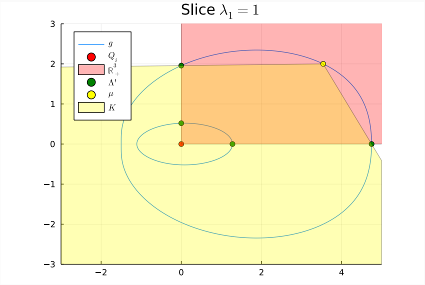
When the spectral curve is hyperbolic, both conditions can be checked by separating convex cones

Redundant Aggregations

Can conclude that $\bigcap_{i=1}^{k} S_{\lambda^{(i)}} = S_\mu \bigcap \left(\bigcap_{i=1}^{k} S_{\lambda^{(i)}} \right)$ by looking at the system

$$\begin{cases} \begin{bmatrix} x^\top & 1 \end{bmatrix} Q_{\lambda^{(j)}} \begin{bmatrix} x^\top & 1 \end{bmatrix}^\top \leq 0, \quad j \in [k] \\ \begin{bmatrix} x^\top & 1 \end{bmatrix} Q_\mu \begin{bmatrix} x^\top & 1 \end{bmatrix}^\top = 0 \end{cases}$$

and counting the number of connected components of $\bigcap_{i=1}^{k} S_{\lambda^{(i)}}$ and $S_\mu \bigcap \left(\bigcap_{i=1}^{k} S_{\lambda^{(i)}} \right)$.



Interlacing Property

Goal: Positive definite certificates of emptiness for restrictions to hyperplanes H

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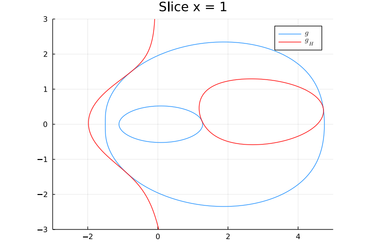
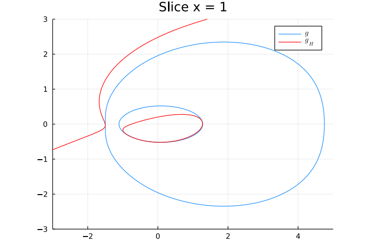
Lemma

Suppose $\mathcal{V}_{\mathbb{R}}(f_1^h, f_2^h, f_3^h) = \emptyset$ and g is smooth and hyperbolic with a hyperbolicity cone $\mathcal{P}$ which contains matrices with exactly two negative eigenvalues. Then, if

$$g_H = \det(Q_\lambda|_H)$$

is smooth and hyperbolic, either $\mathcal{P}_H$ contains positive definite matrices or $\mathcal{P}_H \subseteq \mathcal{P}$.





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Summary

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- We provide non-PDLC cases where

$$\mathrm{conv}(S) = \bigcap_{i=1}^{k} S_{\lambda^{(i)}}$$